GEOMETRIC REDUCEDNESS AT A POINT OF A LOCALLY NOETHERIAN SCHEME SPREADS TO AN AFFINE OPEN NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the two cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. Overlap pre-check: this label occurs in no other staged chapter (run with agent-authoring/stk/card_plan.py varieties before opening).

Allowed standard prerequisites (editorial).

- An affine open neighbourhood whose ring injects into the local ring. Properties, Lemma labelled properties-lemma-ring-affine-open-injective-local-ring (canonical Stacks tag 0BX3); invoked at proof lines 610-611. Supplies the affine open U with R = O_X(U) -> O_{X,x} injective, which is where local Noetherianity of X is used.
- Geometric reducedness at a point in local-ring terms. Same chapter, Lemma labelled lemma-geometrically-reduced-at-point (source line 354, Stacks tag 035W); invoked at proof lines 614-615. Supplies the translation of the hypothesis into geometric reducedness of the local ring O_{X,x} over k.
- A subalgebra of a geometrically reduced algebra over a field is geometrically reduced. Algebra, Lemma labelled algebra-lemma-subalgebra-separable (canonical Stacks tag 030T); invoked at proof lines 617-618. Supplies that R, being a subalgebra of the geometrically reduced ring O_{X,x}, is geometrically reduced, whence U is.

Source reference. Original lemma 0BX3: sources/stacks-chow/tags.txt, source lines 14966–14966. Cross-chapter standard prerequisite invoked once at proof lines 610-611 to obtain the affine open U with an injective map into the local ring; the rest of the argument is retained in the counted span.

Source reference. Original lemma 035W: sources/stacks-varieties/pinned-original-source.tex, source lines 354–354. Same-chapter pointer invoked at proof lines 614-615 for the local-ring characterisation of the hypothesis; the neighbourhood conclusion is retained in the counted span.

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Source reference. Original lemma 030T: sources/stacks-chow/tags.txt, source lines 3960–3960. Cross-chapter standard prerequisite invoked once at proof lines 617-618 for the subalgebra criterion; the passage from R to U is retained in the counted span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Let $x \in X$. Assume X locally Noetherian and geometrically reduced at x. Then there exists an open neighbourhood $U \subset X$ of x which is geometrically reduced over k.*

Proof. Assume X locally Noetherian and geometrically reduced at x. By Properties, Lemma 0BX3 we can find an affine open neighbourhood $U \subset X$ of x such that $R = \mathcal{O}_X(U) \to \mathcal{O}_{X,x}$ is injective. By Lemma 035W the assumption means that $\mathcal{O}_{X,x}$ is geometrically reduced over k. By Algebra, Lemma 030T this implies that R is geometrically reduced over k, which in turn implies that U is geometrically reduced. □